

Mid Semester Examination 2025

Course: B.Tech. (Civil Engineering)

Semester: 3rd

Paper Name: Engineering Mathematics-III

Paper Code: TMA - 302

Time: 1:30 Hours

MM: 50 Marks

Note:

- (1) Answer all the questions by choosing any one of the sub questions.
- (2) Each question carries 10 marks.

Q. 1

(10 Marks)

- (a) Find the Fourier series for $f(x) = x + x^2$ in the range $-\pi < x < \pi$. Hence show that

$$\frac{\pi^2}{6} = 1 + \frac{1}{2^2} + \frac{1}{3^2} + \dots$$

CO-1

OR

- (b) Obtain the Fourier Series for $f(x) = x \sin x$ in $-\pi < x < \pi$.

CO-1

Q. 2

(10 Marks)

- (a) Find the half range Fourier Sine series for $f(x) = x$, $0 < x < 2$

CO-1

OR

- (b) Express the function $f(x) = |x|$ in the range $-\pi < x < \pi$ as Fourier series. Also deduce that

$$1 + \frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \dots = \frac{\pi^2}{8}.$$

CO-1

Q. 3

(10 Marks)

- (a) Find the Laplace transform of the following functions:

(i) $F(t) = e^{-3t} \sin 2t$

(ii) $F(t) = t^2 e^{2t}$

CO-2

OR

(b) Determine the Laplace transform of given function

$$F(t) = \begin{cases} \sin\left(t - \frac{\pi}{2}\right); & t > \pi/2 \\ 0 & ; t < \pi/2 \end{cases}$$

CO-2

Q. 4

(10 Marks)

(a) Find the Laplace transform of the given function

$$F(t) = \frac{\cos(at) - \cos(bt)}{t}$$

CO-2

OR

(b) To show that $\int_0^\infty te^{-3t} \sin t \, dt = \frac{3}{50}$.

CO-2

Q. 5

(10 Marks)

(a) Use Convolution theorem to find the inverse Laplace transform of the given function

$$f(s) = \frac{1}{s^2(s^2+1)}.$$

CO-2

OR

(b) Use Laplace transform method to solve the given initial value problem

$$\frac{d^2y}{dt^2} + 2\frac{dy}{dt} + y(t) = t \text{ given that } y(0) = -3 \text{ and } y(1) = -1.$$

CO-2